

WHO REALLY BENEFITS FROM INDIA'S FAB ECOSYSTEM?

The Hidden Economic Transformation Behind
India's Semiconductor Mission, 2026–2035



GLOBAL REALIGNMENT

India on every
global shortlist



POLICY TO PROGRESS

Real projects.
Real capital. Real
momentum.



DEMAND PULL

Strong domestic
demand driving
long-term growth



MANUFACTURING TRANSFORMATION

Building a resilient,
future-ready
semiconductor
ecosystem



OPPORTUNITY DECADE

2026–2035 will
define India's
semiconductor
future



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The Hidden Economic Transformation Behind India's Semiconductor Mission (2026–2035)

India Semiconductor Intelligence

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KEY JUDGEMENTS

The report's central findings are stated below as analytical judgements rather than as facts. Each carries an explicit confidence rating: High (H) where the supporting evidence is direct and convergent, Moderate (M) where the supporting evidence is partial or where reasonable analysts could reach different conclusions, and Low (L) where the judgement extrapolates beyond the evidence and depends substantially on assumptions about variables that are not yet observable. The reader should treat the report as a working interpretation, not as a forecast, and should track the *signals to watch* listed in Appendix D against these judgements over time.

Judgement 1 — Mature-node sovereignty, not leading-edge competition, is India's correctly calibrated semiconductor objective. (*Confidence: High.*) The combination of domestic demand structure (automotive, defence, telecom, industrial), available technology-transfer pathways (PSMC, Renesas, Micron's own IP), and current geopolitical alignment makes mature-node capacity (28-nm and above logic, advanced packaging, GaN, SiC, analog, power management, sensor) the addressable opportunity for India over the 2026-2035 horizon. Leading-edge competition with TSMC at 3-nm and below is not addressable inside this horizon and should not be the policy objective.

Judgement 2 — Gujarat is the most likely first-wave cluster leader on currently observable evidence, but the lead is not pre-ordained. (*Confidence: Moderate-High.*) Gujarat hosts approximately 80% of approved ISM capex through May 2026, the only operational SIR with sanctioned industrial water and dedicated grid, and the most aggressive state-level incentive stack. The judgement is conditional on (a) the Tata-PSMC Dholera yield ramp through 2026-2028, (b) continuity of Narmada allocation through deficit monsoon years, and (c) Gujarat's continued political-execution bandwidth across electoral cycles. A failure on any of (a)-(c) materially changes the state ranking.

Judgement 3 — Hidden-tier industrial-services firms may capture more cumulative cascade revenue than the anchor chip companies over the 2026-2030 window. (*Confidence: Moderate.*) The judgement is directionally supported by the supplier-disclosure pattern of comparable fab clusters globally (TSMC Hsinchu, Samsung Pyeongtaek, Intel Arizona) and by the early order-book patterns at Tata Projects, INOX Air Products, VA Tech Wabag, Voltas, Blue Star, and Schneider India. The judgement is sensitive to (a) the pace of fab ramp, (b) the depth of localisation that Indian fabs pursue versus continued import dependency, and (c) the relative

margin trajectories of the chip-company portfolio versus the hidden-tier portfolio. The 2028-2030 window will reveal whether the hidden-tier outperformance compounds or reverses.

Judgement 4 — Talent is the binding constraint of the entire mission, and the constraint cannot be dissolved by capital or policy alone. (*Confidence: High.*) India's current ~220,000 semiconductor professionals are concentrated in design; the manufacturing-tier pool (process integration, yield engineering, equipment maintenance, advanced packaging) is structurally thin and cannot be trained to operational competence in less than 36-60 months. The 250,000-350,000 person gap projected for 2027 is the bookkeeping definition of the talent constraint and is, in our reading, the variable most likely to bind across all three scenarios in Chapter 12.

Judgement 5 — Semiconductor outcomes depend more on ecosystem density than on fab count. (*Confidence: High.*) Taiwan's industry is not TSMC but the ~20,000-firm supplier cluster around it. Korea's industry is not Samsung but the ~12,000-firm cluster. India's destiny is the cluster pattern that develops around the anchor fabs and OSATs, not the anchor count itself. The state-by-state, corridor-by-corridor cascade analysis in Chapters 6-8 is therefore the operationally consequential analysis. The fab approval count is a leading indicator; the cluster density is the outcome variable.

METHODOLOGY AND ANALYTICAL FRAMEWORK

This section sets out the analytical method, sourcing hierarchy, projection assumptions, and explicit limitations of the report. It is included before the executive summary so the reader can calibrate the strength of the report's claims against its evidentiary base before engaging with the substantive analysis.

Sourcing Hierarchy

The report's evidentiary base draws from four tiers of source material, in descending order of weight.

Tier 1 (highest weight): Primary disclosures. These include Cabinet approvals and Press Information Bureau (PIB) releases; the Ministry of Electronics and Information Technology (MeitY) policy notifications; the India Semiconductor Mission (ISM) project list and status updates; state-level semiconductor policy notifications for Gujarat (2022-27), Uttar Pradesh (2024), Karnataka (ESDM 2023-28 and Semicon 2024), Tamil Nadu (Semiconductor and Advanced Electronics Policy 2024), Telangana (Electronics Policy 2.0); and the public stock-exchange disclosures of listed firms named in the report.

Tier 2: Corporate disclosures by international counterparties. These include the technology-transfer partner statements (PSMC, Micron, Renesas, ASML, Tokyo Electron, Applied Materials, Lam Research, KLA), the global foundry capacity and revenue disclosures (TSMC, Samsung, SK Hynix, UMC, GlobalFoundries), and the materials and equipment supplier disclosures (Tokyo Ohka Kogyo, JSR, Shin-Etsu, Fujifilm, Cabot Microelectronics, DuPont, Linde, Air Products, Air Liquide).

Tier 3: Industry and trade-press reporting. These include the trade-press coverage by Communications Today, Indian Chemical News, Business Standard, Deccan Herald, BusinessToday, Outlook Business, ThePrint, Tribune India, Techovedas, EE Times, Cleanroom Technology, gasworld, SemiWiki, Digitimes, Tom's Hardware, Light Reading, and similar publications; data from industry associations (IESA, SEMI India, ELCINA, ESSCI, India Electronics and Semiconductor Association); and analyst-firm publications (Deloitte, McKinsey, BCG, MarketsandMarkets, Mordor Intelligence).

Tier 4: Modelled estimates. These include the cascade-revenue projections, the MSME niche count, the cluster-employment estimates, the cumulative supplier-capex projections, and the probability-weighted scenario expected values. These are explicitly identified in the text where they appear and are accompanied by the assumption set underlying them.

The cumulative claim of the report is that approximately 70% of the named numerical assertions are Tier 1 or Tier 2 sourced; approximately 20% are Tier 3 sourced; and approximately 10% are Tier 4 modelled estimates. Where the modelled estimates appear in the text, the basis is identified ("derived from", "modelled using", "benchmark comparison").

Projection Methodology

The report's quantitative projections rest on a small number of explicit assumptions. The reader should test the projections against alternative assumption sets.

Cascade multiplier (the "500+ industries per fab" claim). Derived from the supplier-disclosure patterns of TSMC (Hsinchu Science Park supplier directory, approximately 3,200 vetted suppliers across the cluster, of which approximately 600 are first-tier), Samsung Pyeongtaek (approximately 1,800 first-tier suppliers documented in published Korean industrial data), and Intel Arizona (CHIPS Act supply-chain disclosures filing approximately 700 first-tier suppliers). The 500-1,000 supplier figure for a single mature-node fab is the median across these three reference clusters, scaled to the Tata-PSMC Dholera fab spec. Confidence: Moderate (the multiplier is observable but cluster-specific variance is large).

Cumulative beneficiary firm count (150-250 firms). Derived from the published listed-firm filings of approximately 100 named companies across the eight tiers in Chapter 8, augmented by industry-association membership rolls (IESA approximately 280 members, ELCINA approximately 700 members, ESSCI training-partner roster approximately 150 institutions) and unlisted private-firm disclosures where available. The 150-250 range reflects the uncertainty in attributing meaningful business impact to firms with marginal cascade exposure. Confidence: Moderate.

MSME qualified-supplier count (4,000-8,000 by 2035). Derived from the supplier-base density patterns of Vietnamese, Malaysian, and Thai electronics clusters (which have built MSME populations in the 5,000-15,000 range over comparable timeframes from comparable bases), scaled down by India's slower expected qualification cadence in the first decade. Confidence: Low-Moderate (the figure is sensitive to AVL throughput and to the depth of localisation that fabs and OSATs pursue).

Direct employment (600,000-900,000 by 2035). Derived from the IESA workforce projections (470,000-570,000 by 2027 demand-side) extended to 2035 with assumed cumulative annual growth of 6-8% in semiconductor and adjacent employment. Confidence: Moderate.

Domestic semiconductor market (\$80-250 billion 2035 range). Derived from Deloitte and IESA forecasts (\$100-120 billion by 2030 central case) extended to 2035 with scenario-dependent CAGR assumptions of 8% (Conservative), 12% (Accelerated), and 18% (Break-through). Confidence: Moderate (consistent with multiple independent forecaster ranges).

Hidden-tier cumulative capex (\$25 billion through 2030). Derived from announced EPC contract values, INOX/Linde plant capex disclosures, VA Tech Wabag and Ion Exchange UPW project estimates, Voltas/Blue Star HVAC pipeline guidance, and scaling of the per-fab utility-capex ratio (approximately 25-30% of fab capex spent on adjacent utility and services infrastructure). Confidence: Moderate.

The reader should note that the report's projections are *directional* rather than precise. The intent is to bound the range of plausible outcomes rather than to forecast specific values; the value of these projections lies in their *relative* magnitudes (cascade > anchor; talent > capex; hidden tier > chip P&L through 2028) more than in their absolute levels.

Definition of "Beneficiary"

The report's beneficiary framework requires a definition. A firm or category is treated as a beneficiary if (a) it has documented direct revenue, contract, or order-book exposure to the semiconductor mission, OR (b) it operates in an industrial category where the cascade pattern observable globally indicates a measurable revenue uplift through 2030, AND (c) the firm has the operational capacity to convert the exposure into realised revenue within the report's horizon. The eight-tier framework in Chapter 8 distributes firms across this definition, with Tier 1 highest-direct and Tier 8 most-indirect.

The framework does *not* treat as beneficiaries firms whose semiconductor exposure is purely demand-side (i.e., firms that *consume* chips, like consumer-electronics OEMs); the cascade benefit accrues to upstream and adjacent firms, not to downstream consumers. Tier 7 (automotive) and Tier 8 (data centre, telecom) appear in the framework because these firms also drive demand pull for domestic chips, which feeds back into the cascade through localisation incentives, but their beneficiary status is genuinely secondary to the upstream tiers.

Confidence Rating System

Throughout the report, major analytical claims are accompanied by an explicit confidence rating:

- ****High (H)****: direct evidence, convergent across multiple Tier 1 or Tier 2 sources; unlikely to be substantially revised on new information.
- ****Moderate (M)****: partial evidence; reasonable analysts could reach different conclusions; sensitive to assumptions that are stated.
- ****Low (L)****: extrapolation beyond available evidence; depends on variables that are not yet observable; should be treated as working hypothesis only.

The ratings are author judgements rather than formal statistical confidence intervals. The intent is to enable the reader to weight the report's claims appropriately rather than to assert a level of precision the analysis does not have.

Limitations

The report has six explicit limitations the reader should hold in mind.

First, the projection horizon (2035) extends beyond reliable forecasting capacity. Forecasts ten years out depend on policy, geopolitical, and demand variables that cannot be predicted with confidence. The scenarios in Chapter 12 are designed to bound the range; they are not predictions of which scenario will obtain.

Second, the report's cascade and beneficiary claims rest on the assumption that the operational fabs and OSATs ramp on schedule or with manageable slippage. If the Dholera yield ramp fails materially through 2028 (the report assigns approximately 15% probability to a material failure scenario), the cascade pattern will be substantially smaller than projected.

Third, the report's geopolitical assumptions favour continued US-India strategic alignment and continued favourable technology-transfer access. A material drift in either variable would tighten the operational environment significantly and is not separately scenario-modelled.

Fourth, the report does not contain proprietary corporate or financial data that would be available in a paid institutional research subscription. The corporate identification is at the named-firm level; the financial implications are directional rather than calibrated to specific earnings or valuation models.

Fifth, the MSME niche identification is based on global supplier-base patterns adapted to Indian industrial conditions; the niche viability for any specific entrepreneur depends on local circumstances (state, capital access, technical capability, AVL relationships) that the report cannot evaluate.

Sixth, the report's analysis is current to May 2026. Material events after this date (new ISM approvals, policy changes, geopolitical shifts, demand surprises, execution slippage) will, by definition, not be reflected. The report is intended to be re-issued on a regular cadence and is best read in conjunction with the Signals to Watch list (Appendix D).

The reader who finds these limitations more demanding than expected should regard the explicit acknowledgement of them as evidence of analytical seriousness rather than as a weakness.

EXECUTIVE SUMMARY

The Thesis in One Page

India will not build a Taiwanese semiconductor industry. It will build something stranger, more diffuse, and arguably more durable: a semiconductor *ecosystem* that distributes industrial gains across roughly seven states, two hundred and fifty Indian companies, and an unknown but very large number of MSMEs whose names will not appear in newspaper coverage. The two operational fabs (Tata-PSMC Dholera entering wafer-out in late 2026 to 2027, and the OSAT cluster at Sanand already producing in 2026) are necessary infrastructure, not the destination. The destination is the cascade.

The cascade is the simple proposition that every fab requires a population of suppliers larger than itself, and that almost none of those suppliers are visible in the semiconductor narrative. A 50,000-wafer-per-month foundry at 28 nm consumes between 20,000 and 30,000 cubic metres of raw water per day, recycles roughly 75 to 85 per cent of it, runs continuous high-purity nitrogen and oxygen lines that demand a captive industrial-gases plant, operates inside cleanrooms whose construction is contracted to a small number of EPC firms with the right cleanroom-class experience, sits on industrial real estate that revalues neighbouring land by multiples, draws its base load from a captive power architecture that the public grid cannot natively support, and pays its wage bill into a regional labour market that did not exist before the fab arrived. The fab is the anchor. The cascade is the economy.

This report identifies the cascade explicitly. Direct first-wave winners — OSATs, EMS firms, cleanroom builders, industrial-gas producers, ultra-pure water EPC, ESD ecosystems, packaging vendors, automated test, factory automation — are the visible top of the pyramid. Beneath them, the hidden tier includes industrial real estate trusts, Tier-2 city service economies, specialty HVAC, precision power conditioning, hazardous-chemical waste handling, NABL-accredited calibration labs, gas-cylinder reconditioning, semiconductor-grade logistics, and a training-and-certification market growing on the back of a 250,000–350,000-person skill gap. Beneath the hidden tier sits the MSME tier, where roughly two hundred discrete niches exist with entry capex between ₹20 lakh and ₹10 crore. These niches are the report's central operational finding.

Who Wins — In One Sentence Each

Gujarat is, on currently observable evidence, the most likely first-wave cluster leader; the gap relative to other states is expected to widen through 2027 if the Dholera ramp holds (Confidence: Moderate-High). Dholera, Sanand and Surat have become India's first true semiconductor cluster, anchored by Tata-PSMC, Micron, Kaynes, CG Semi-Renesas, Crystal Matrix and Suchi Semicon, supported by the most aggressive state subsidy stack in the country and the only large greenfield industrial region with sanctioned Narmada water and a dedicated grid.

Tamil Nadu appears positioned to consolidate the EMS layer but is unlikely to land a front-end fab inside this decade (Confidence: Moderate) — Foxconn, Tata Electronics, Dixon, Pegatron and Wistron now anchor an electronics export base above 41 per cent of the national total, but the absence of fab-grade greenfield land with allocated ultra-pure water means the state's semiconductor position is the *output* side rather than the wafer side.

Karnataka retains the design layer but, on present evidence, has not yet translated design strength into manufacturing presence (Confidence: High on design, Moderate on the manufacturing-gap interpretation) — Bengaluru remains the largest chip-design city in India and one of the largest globally (AMD, Intel, Qualcomm, Nvidia, NXP, AMD, Texas Instruments all maintain India headquarters here), but a decade of design dominance has not translated into a single approved fab, and the gap between design strength and manufacturing-utility strength is the state's defining problem.

Telangana holds the captive-design layer (Qualcomm's second-largest global campus, MosChip, AMD design centres) and the AI/data-centre tier, but lost the front-end fab race to Gujarat because it could not assemble a single greenfield SIR with allocated water, dedicated grid, and a state capex top-up of the scale Gujarat offered.

Uttar Pradesh is the most credible dark-horse cluster candidate — HCL-Foxconn at Jewar, the new Noida International Airport, the YEIDA cluster, and a 50 per cent capital subsidy on top of ISM make UP the second most aggressive state policy after Gujarat; if UP executes through the 2026–2030 window, it is the most plausible second cluster after Gujarat.

Assam has anchored the OSAT layer for the Northeast — Tata's ₹27,000 crore Jagiroad facility entering Phase 1 commissioning in April 2026 will produce up to 48 million chips per day at full ramp, and it has anchored Northeast India's first serious manufacturing transformation since independence.

Maharashtra is the unexpected first-cycle laggard — the Adani-Tower \$10-billion fab approved by the state in September 2024 collapsed at the ISM stage by mid-2025; the state will need to rebuild credibility through fresh anchor announcements.

What Is Hiding in Plain Sight

A short list of companies that almost no general-interest media coverage of India's semiconductor programme has identified as central beneficiaries, but that the operational evidence places at the centre of the cascade: **INOX Air Products** (₹500-crore Dholera specialty-gas plant supplying the Tata-PSMC fab), **Linde India** (parallel Dholera gases plant), **VA Tech Wabag** (ultra-pure water lifecycle partner, the largest opportunity in industrial water in India this decade), **Tata Projects** (built Micron Sanand and is building Tata-PSMC Dholera — the dominant EPC for cleanroom-grade construction), **Voltas** and **Blue Star** (precision HVAC for fabs and data centres), **Navin Fluorine** and **Gujarat Fluorochemicals** (electronic-grade fluorochemicals), **Schneider Electric India** and **Vertiv** (power conditioning and UPS at fab scale), **Cummins India** (backup power), **Embassy REIT** and **Mindspace REIT** (industrial real estate that will be re-valued by clusters), **3M India**, **Kimberly-Clark** and **DuPont India** (ESD and cleanroom consumables, currently 100 per cent imported and overdue for substitution), **Cyient DLM** (ASICs and adjacent EMS), **NIELIT**, **IISc-Synopsys ISWDP** and **ESSCI** (the skilling layer monetizing the 85,000-engineer ISM target).

On the supplier-disclosure pattern observable in comparable fab clusters globally (Hsinchu, Pyeongtaek, Arizona) and on the early order-book evidence in India through May 2026, these twelve names are positioned to see a cumulative revenue impact from the fab cascade through 2030 that, in our estimate, is comparable to — and may exceed — the impact accruing to several of the directly approved chip companies. (Confidence: Moderate; sensitive to fab ramp pace.)

The Binding Constraints — Stated Plainly

India's semiconductor mission has three binding constraints that no amount of capital or policy can resolve in the 2026–2030 window. The water problem is real but solvable with capex (Narmada allocation, ZLD plants, 90 per cent recycle targets); the power-quality problem is real and harder (the Indian grid does not natively meet SEMI F47 ride-through, sub-cycle voltage sags scrap wafer lots, and Gujarat's grid frequency breached the IEGC ± 0.05 Hz band repeatedly through 2023–2025); the talent problem is the binding one — India has roughly 220,000 semi-

conductor professionals concentrated in design, against a forecast 470,000–570,000 requirement by 2027, with the entire gap concentrated in process integration, yield engineering, equipment maintenance and advanced packaging, none of which can be trained in less than three years. The fourth structural constraint — and the one this report treats with the most realism — is the equipment and materials monopoly: ASML on lithography, Tokyo Electron and Applied Materials and Lam Research on deposition and etch, KLA on metrology, Shin-Etsu and SUMCO on silicon wafers, and the Japanese photoresist quartet (TOK, JSR, Shin-Etsu, Fujifilm) controlling roughly 80 per cent of global photoresist and over 95 per cent of EUV resist. None of these positions can be substituted by India this decade. India's strategic move is not substitution; it is *qualified diversification* — becoming the credible third or fourth geographic node for non-EUV nodes (28, 40, 55, 65, 90, 110 nm) where global demand is rising faster than installed capacity.

Three Scenarios to 2035

Conservative (60 per cent probability): India ends 2035 with 5–8 per cent of global mature-node foundry capacity, \$80–100 billion semiconductor market, around 600,000 direct semiconductor and adjacent jobs, OSAT exports at \$15–20 billion, and continued dependence on imported leading-edge logic, EUV-class equipment, and most materials. Industrial cascade is real but uneven; Gujarat is the only state-level success story; the MSME ecosystem reaches roughly 8,000 qualified suppliers.

Accelerated (30 per cent probability): India enters the global club. 12–15 per cent of mature-node foundry capacity, \$150 billion market, 900,000 direct jobs, a third operational fab approved, and a credible 22 nm or 16 nm capability through technology transfer. Gujarat plus one or two other states (UP, TN-via-fab-late, or Karnataka-via-design-to-fab) reach cluster scale.

Breakthrough (10 per cent probability): India becomes the global alternate. 20 per cent of mature-node capacity, \$250 billion market, indigenous equipment in select categories (metrology, gas delivery, automation), and the first credibly Indian-led advanced-packaging capability at scale. This scenario requires policy continuity through three political cycles, a step-change in talent throughput, and at least one indigenous champion at the OSAT-to-fab transition that does not yet exist.

The probability-weighted expected outcome — India at roughly 9–10 per cent of mature-node capacity, \$115–125 billion market, and 600,000–900,000 direct semiconductor and adjacent jobs (the spread reflects scenario uncertainty) — would, if realised, constitute the most significant

manufacturing-led industrial transformation in India since the post-1991 services liberalisation (Confidence: Moderate, conditional on cluster ramp). (Figure 1)



Figure 1 — Strategic timeline 2026–2035 — milestones across fabs, OSATs, design, skilling, policy, and demand swim lanes.

What This Report Recommends, in Six Sentences

For MSMEs: pick a niche in the Low-Difficulty / High-Profitability quadrant of the Chapter 7 matrix, begin AVL qualification at Sanand or Jewar within the next six months, and target first revenue inside fifteen months. For investors: prefer the hidden-tier industrial-services basket (water, gases, HVAC, EPC, calibration, RE) over the headline chip names for the 2026–2028 window; the cascade compounds before the chip P&L does. For state governments: water sanction, dedicated grid, and a 75 per cent land-and-stamp-duty subsidy stack are now the table-stakes; the differentiated state will compete on power quality, technician throughput, and ESG transparency. For large enterprises: the OSAT-to-fab transition is the strategic acquisition window — the next decade will produce two to four Indian semiconductor champions, and the price is lowest before 2028. For academia: the largest commercial market in education India will see this decade is process and equipment training; design training is over-supplied. For policymakers: ISM

3.0 must shift incentive weight from capex toward materials and equipment substitution; capex incentives have done their job.

The report's underlying claim is that the semiconductor mission, viewed as an *industrial transformation* rather than a chip programme, has already succeeded in the sense that matters: it has created a credible, sanctioned, capitalised industrial geography in five states and has begun the slow accumulation of supplier ecosystems around it. The unfinished work is everything else.

ter trajectory is, on this comparison, neither faster nor slower than the Korean precedent; it is approximately on-track.

What India Has That Taiwan, Korea and Malaysia Did Not

For the institutional reader who needs the analytical frame compactly:

(a) **Larger domestic demand pull at start.** India enters its semiconductor build-out with a \$45-50 billion domestic semiconductor market in 2026 — substantially larger than Taiwan's domestic market in 1987 (\$3 billion) or Korea's in 1985 (\$2 billion). The demand pull is the structural feature that gives India a different industrial-logic than the predecessor cluster countries had at comparable points.

(b) **A pre-existing design ecosystem.** Bengaluru and Hyderabad were already among the world's largest chip-design cities at ISM launch. Taiwan and Korea had to build the design layer concurrently with the manufacturing layer; India is doing the opposite, building manufacturing into a country that already has the design layer.

(c) **A favourable geopolitical window.** Taiwan and Korea's industries grew during a period when the United States and Japan welcomed allied capacity expansion at the leading edge; China's industry grew despite progressively tightening US export controls. India's industry is being built during a period when the United States and Japan are actively diversifying out of single-jurisdiction concentration risk and are supportive of allied capacity expansion at non-leading-edge nodes. This is a more favourable window than China had and is roughly comparable to Korea's 1985-1995 window.

What India Lacks That Taiwan, Korea and Malaysia Have

(a) **Process engineering depth.** Taiwan's process-engineering pool was 30 years in the making; India's is starting from a small base. This is the binding workforce constraint.

(b) **A unified state capacity equivalent to ITRI.** Taiwan's Industrial Technology Research Institute was the unified state capacity that anchored the technology transfer, talent training, and ecosystem development. India's equivalent is distributed across IISc, IITs, NIELIT, C-DAC, MeitY, and state agencies, with no single institution carrying ITRI's operational scope.

(c) **Equipment and materials base.** Taiwan, Korea, Japan, and now the EU and US have ecosystems of domestic equipment and materials suppliers. India has effectively none in equipment

and a small base in select chemicals. The equipment-and-materials gap is the longest substitution horizon in the report.

(d) **Operational fab cluster maturity.** Pyeongtaek's first Samsung fab dates to 1996; Hsinchu's first TSMC fab to 1987. The cluster maturation takes decades. India's clusters are at year 4-5; they need another 15-20 years to reach the operational maturity of the established clusters.

The report's central position is that the gaps are real but addressable over 15-20 years, the advantages are real and meaningfully large, and the trajectory matters more than the snapshot. India in 2035, on the central scenario, will be roughly where Korea was in 2005 — a credible, mid-sized, partially-sovereign semiconductor industry that is not the global leader but is a structural participant in the global ecosystem. That is, by any historical measure, a substantial achievement.

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